Data Mining and Neural Networks

# Tree Ensembles

Week 03

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MA3022/MA4022/MA7022

School of Computing and Mathematical Sciences
University of Leicester

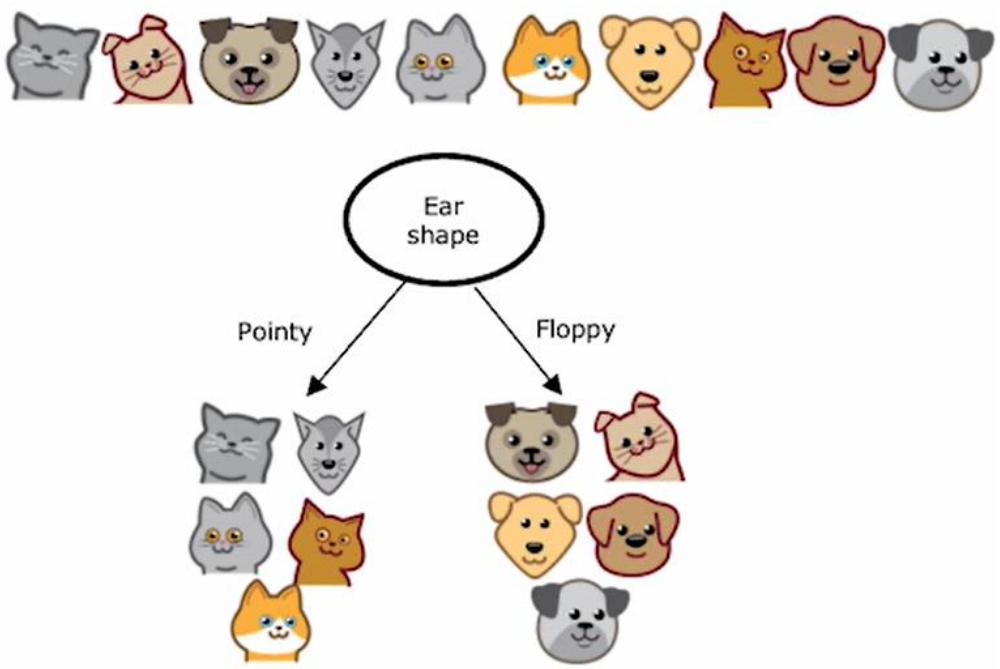
27/28/30-Jan-2026

Credits: Andrew Ng

# Agenda

- Decision trees
  - **Tree ensembles**

# Trees are highly sensitive to small changes of the data



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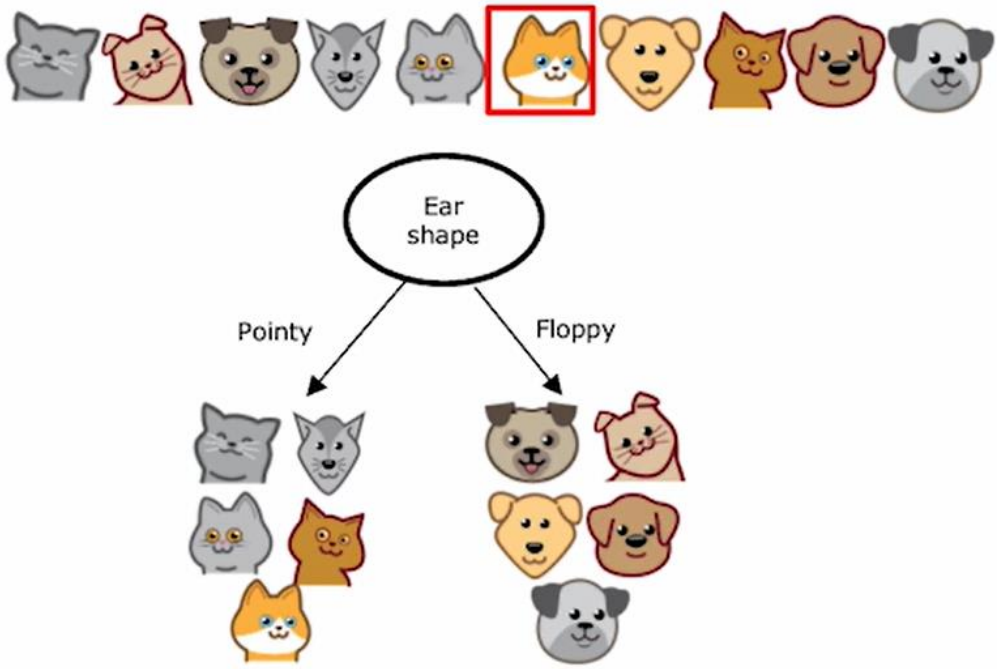

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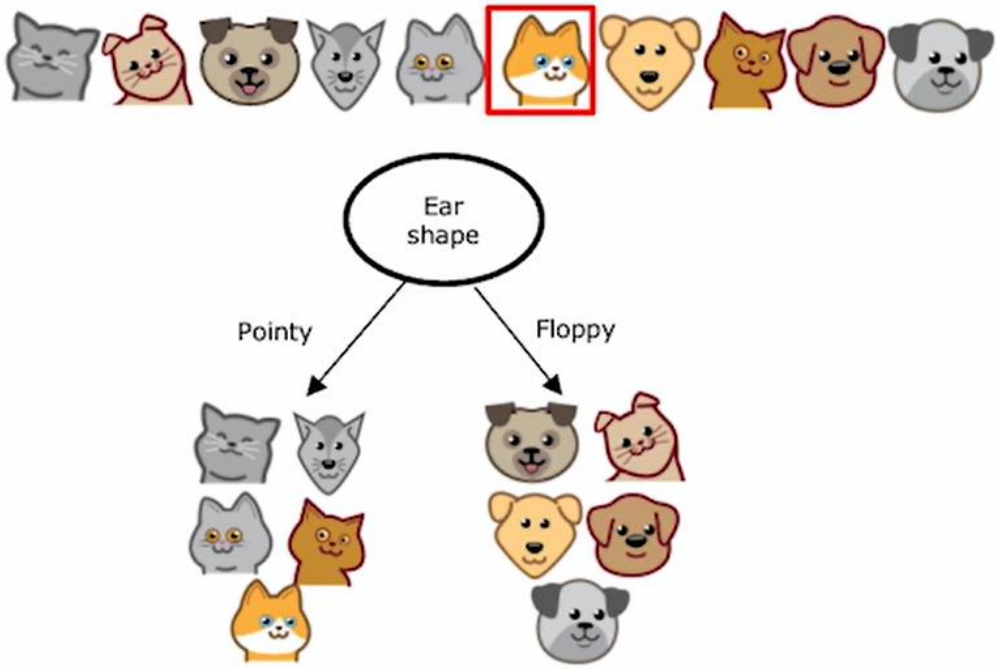


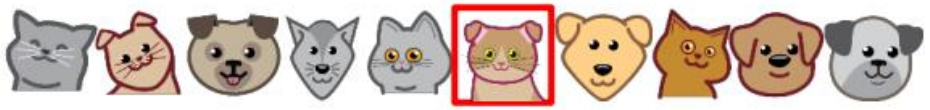

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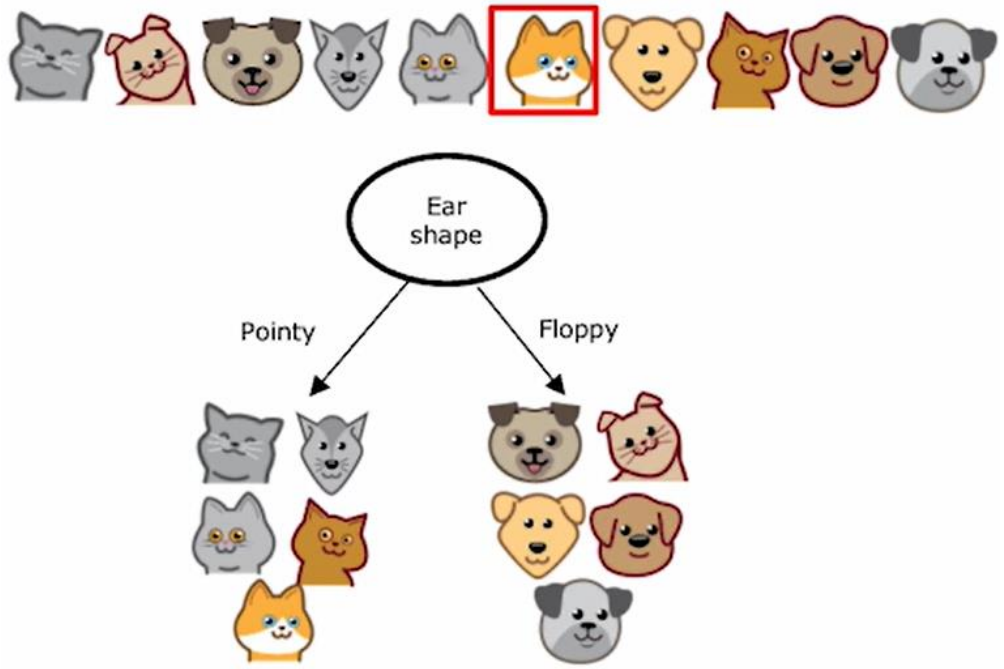
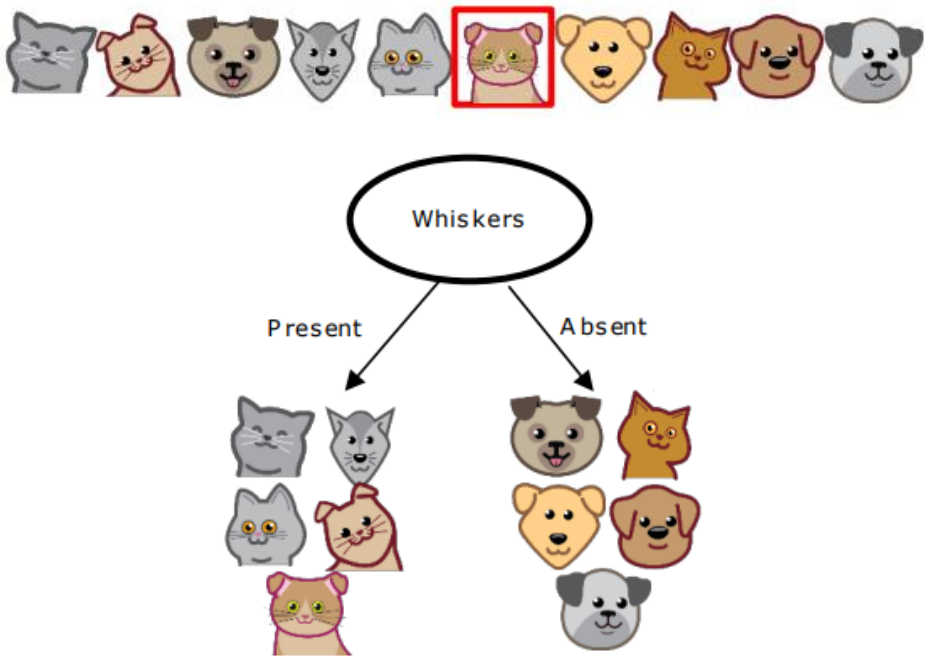
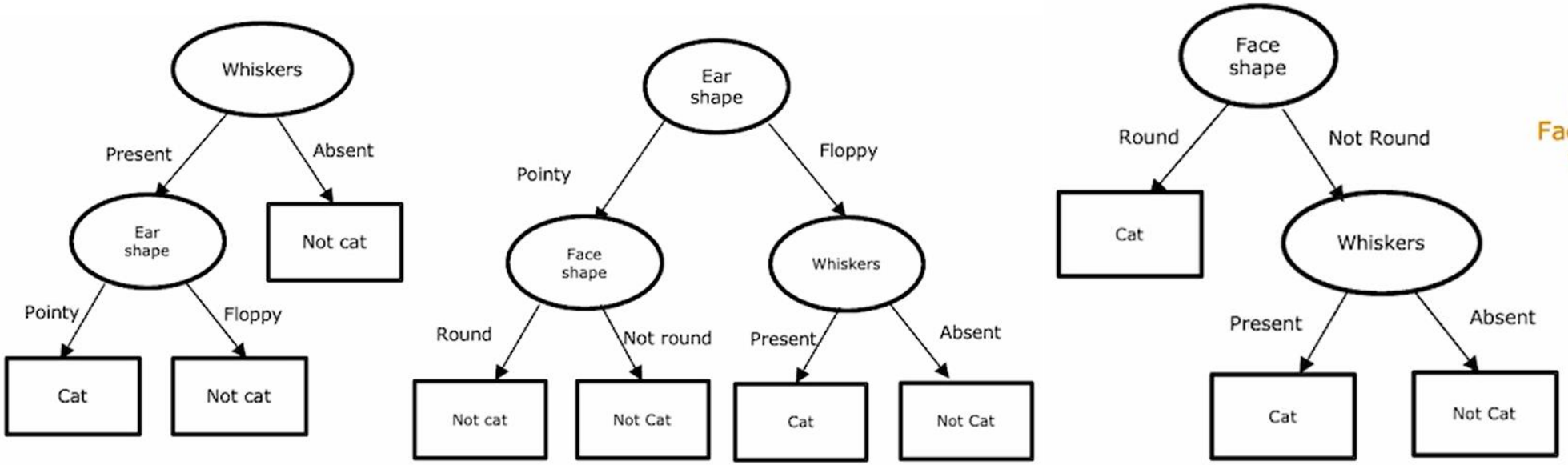


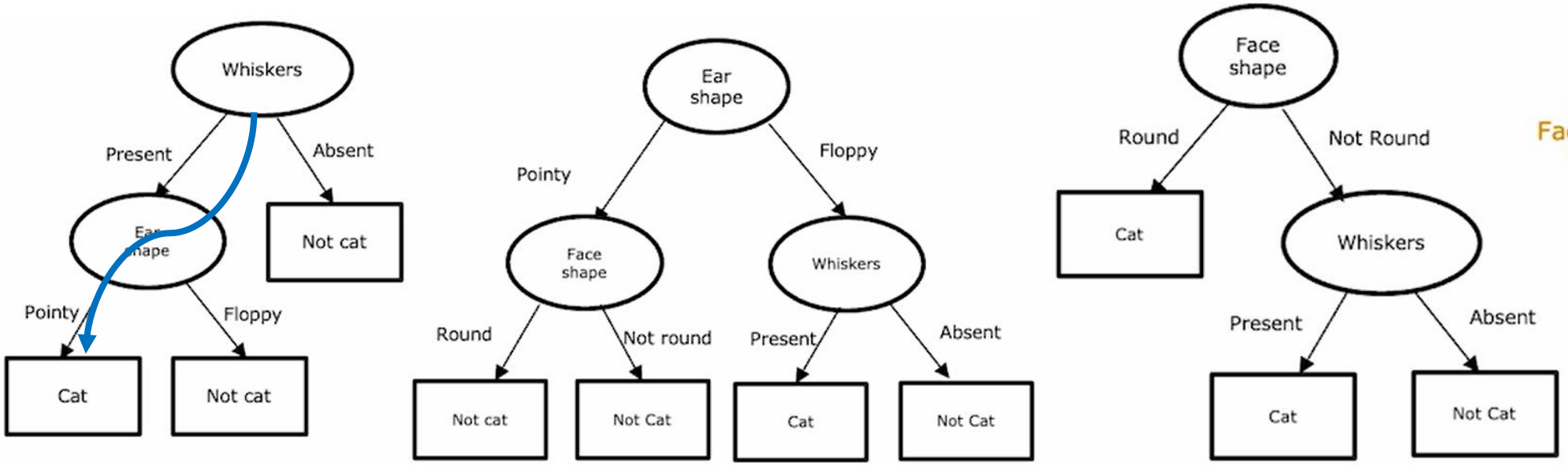
# Tree Ensemble




New test example

Ear shape: Pointy
Face shape: Not Round
Whiskers: Present

# Tree Ensemble

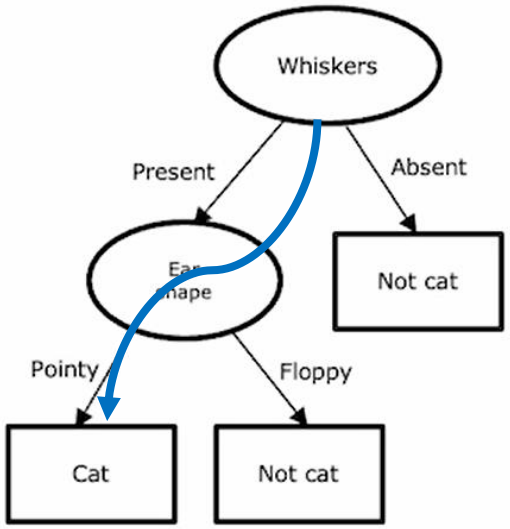


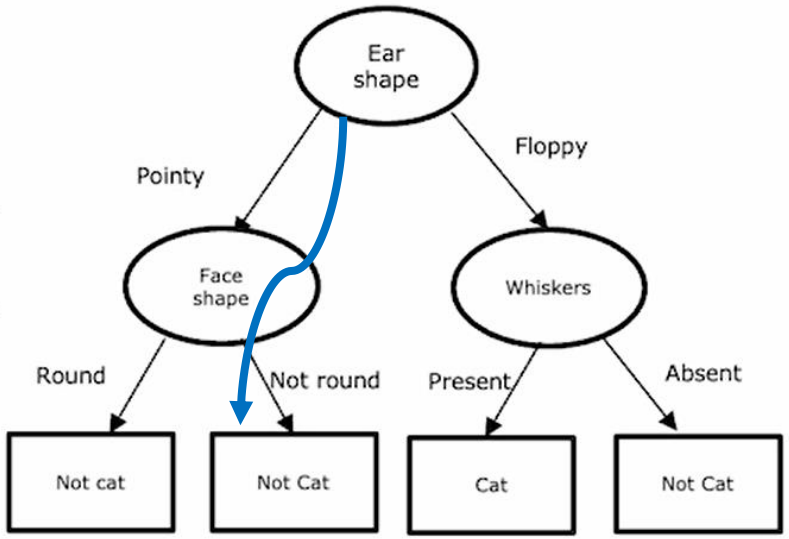

New test example

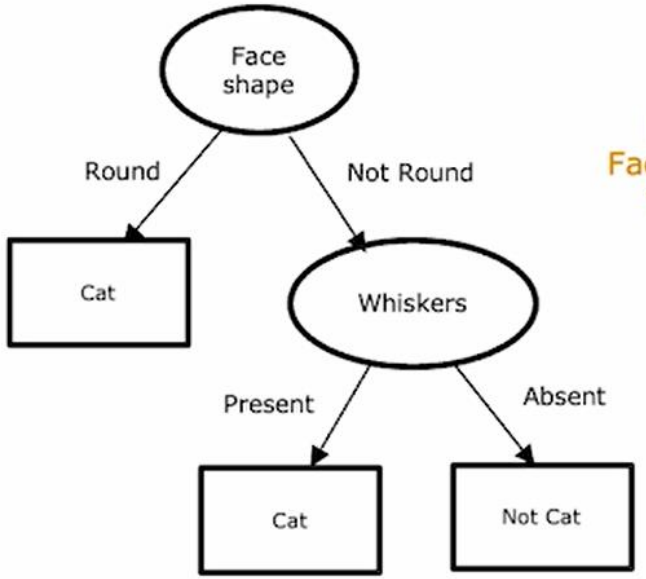
Ear shape: Pointy
Face shape: Not Round
Whiskers: Present

Prediction: Cat

# Tree Ensemble



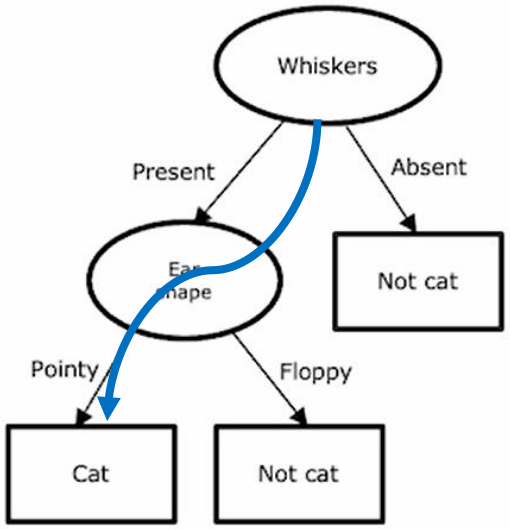

New test example

Ear shape: Pointy
Face shape: Not Round
Whiskers: Present

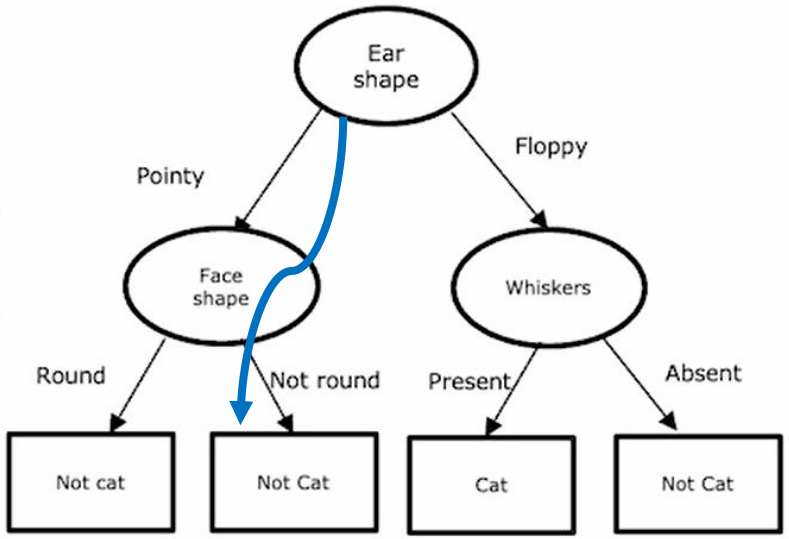
Prediction: Cat

Prediction: Not cat

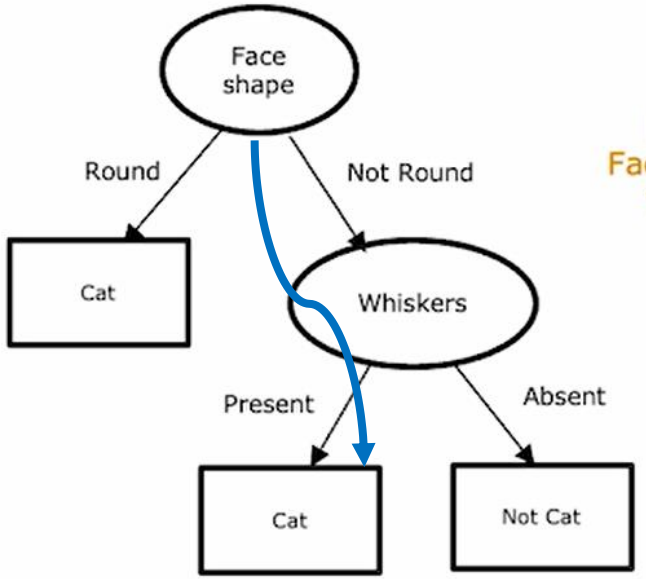
# Tree Ensemble




Prediction: Cat




Prediction: Not cat




Prediction: Cat

New test example

Ear shape: Pointy
Face shape: Not Round
Whiskers: Present

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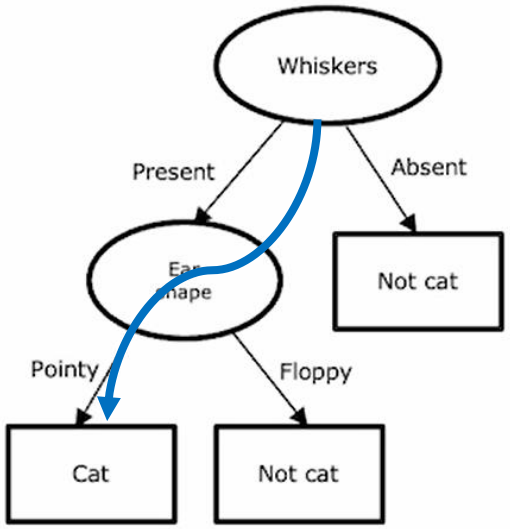
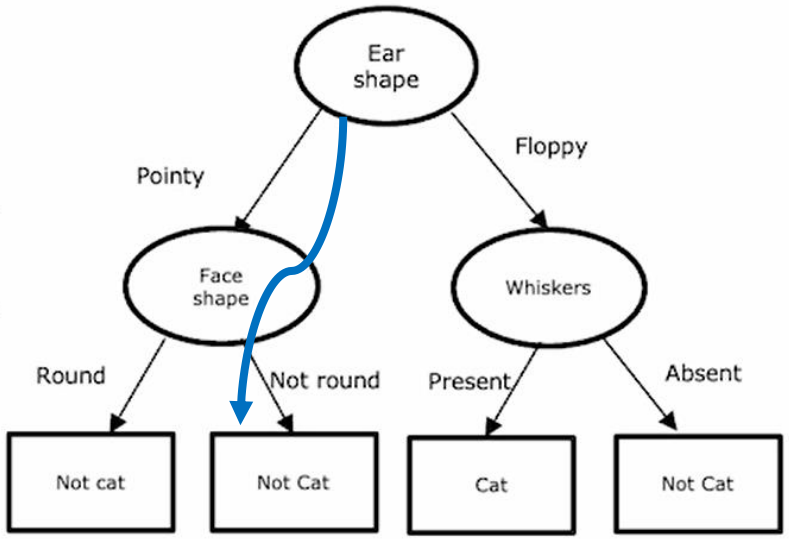
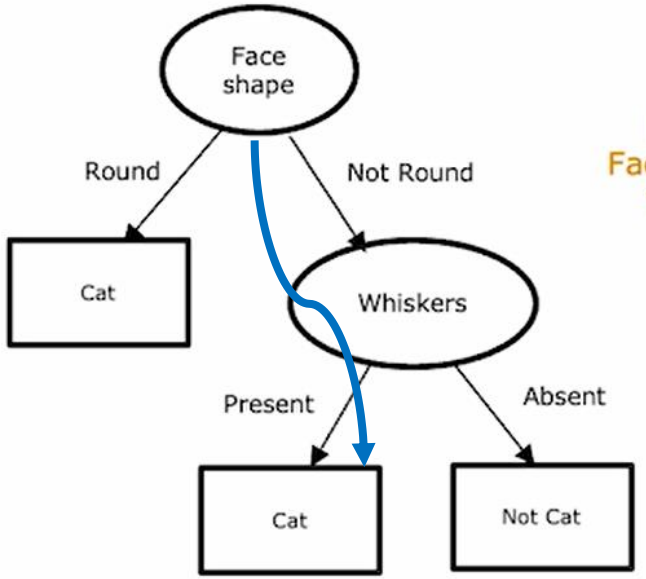
# Tree Ensemble




Prediction: Cat




Prediction: Not cat




Prediction: Cat

New test example

Ear shape: Pointy
Face shape: Not Round
Whiskers: Present

Final prediction: Cat

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# Agenda

- Decision trees
  - Tree ensembles
  - **Sampling with replacement**

# Sampling with Replacement

Tokens

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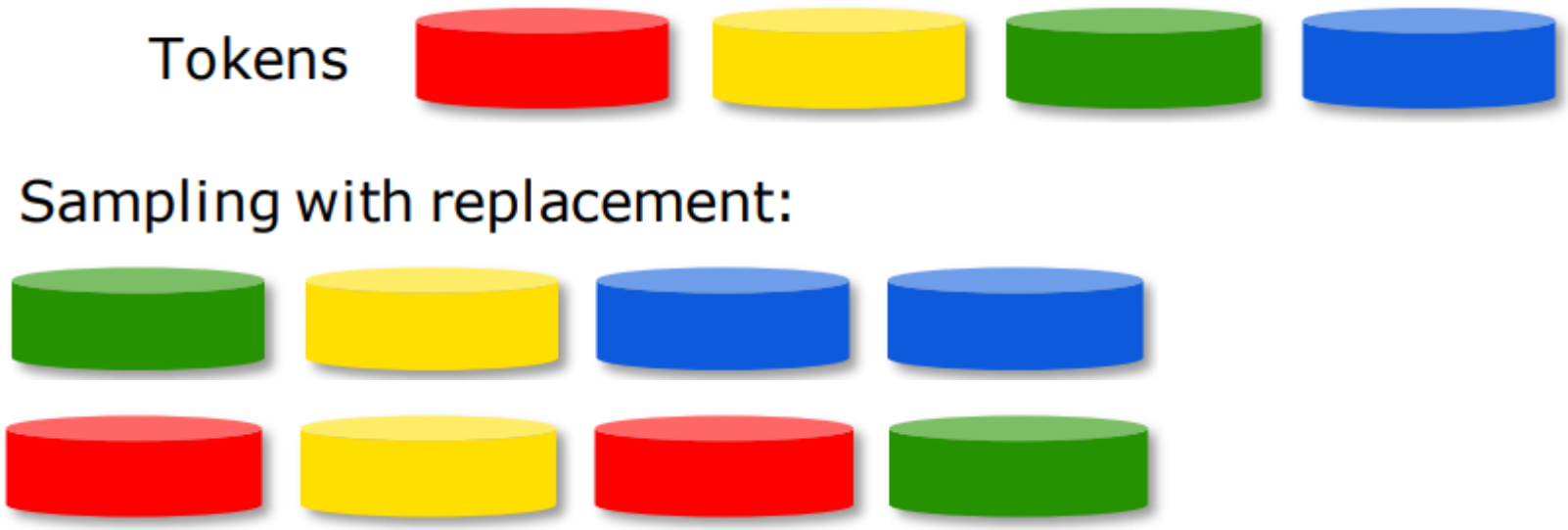
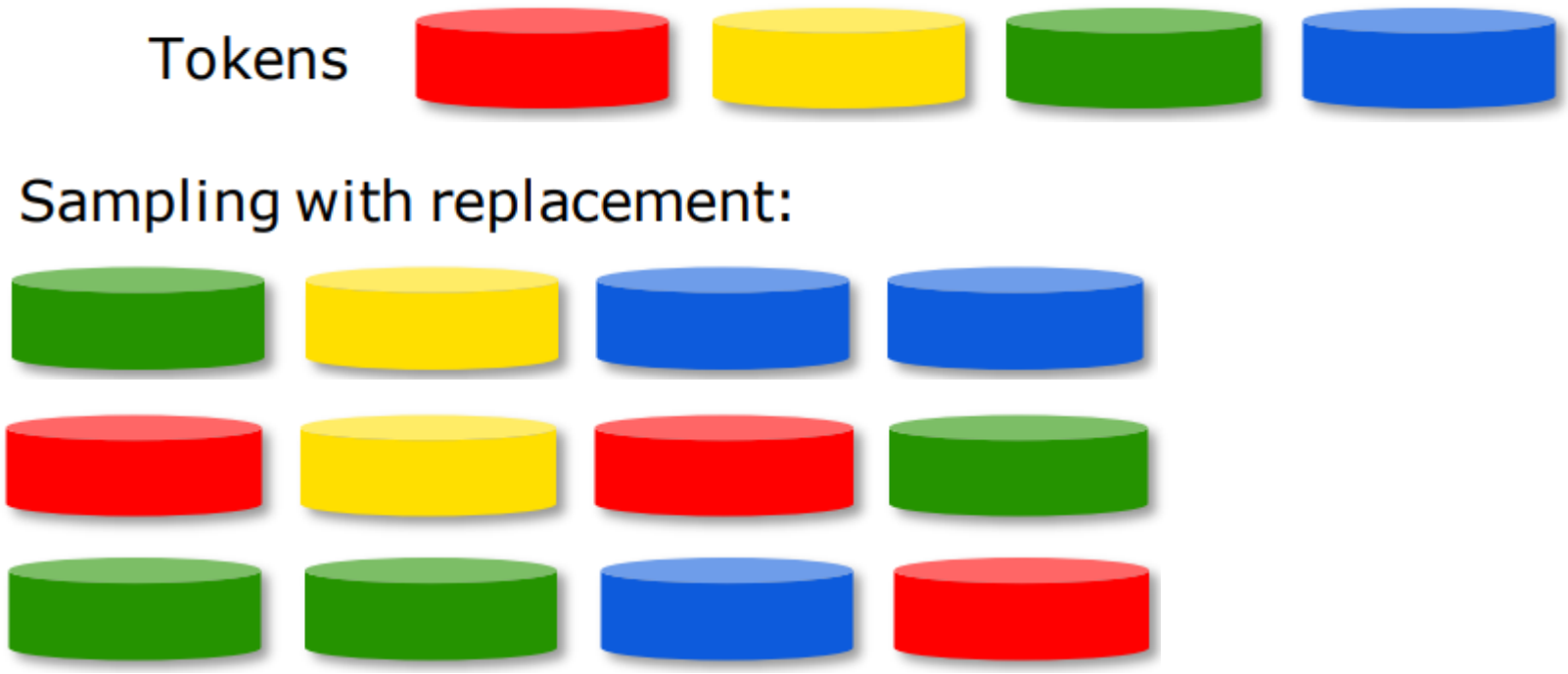
Tokens

Sampling with replacement:

# Sampling with Replacement



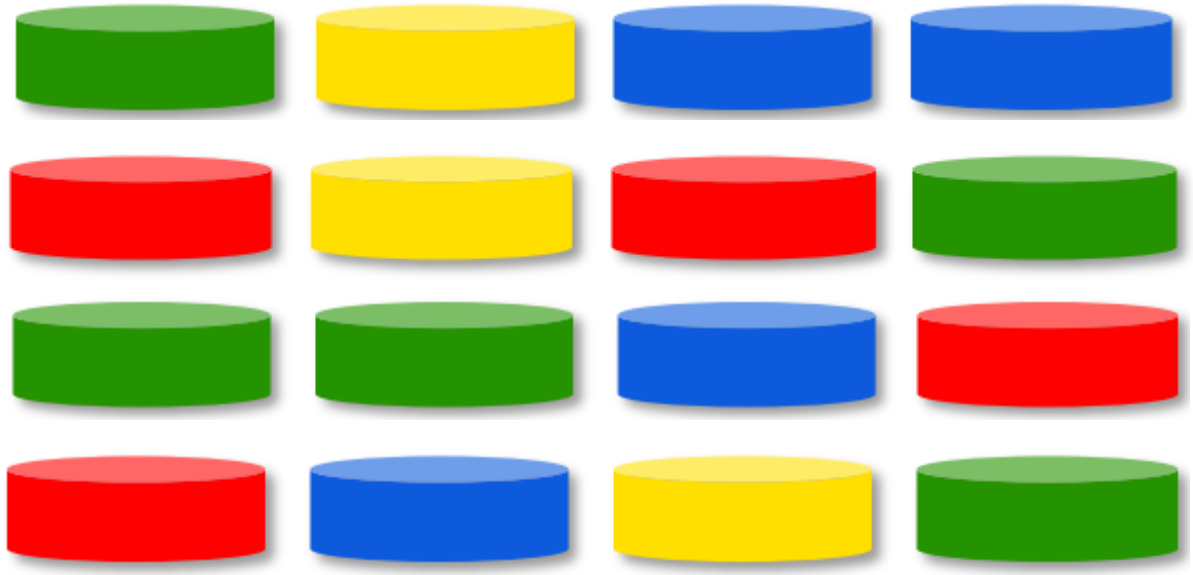
# Sampling with Replacement



# Sampling with Replacement

Tokens

Sampling with replacement:



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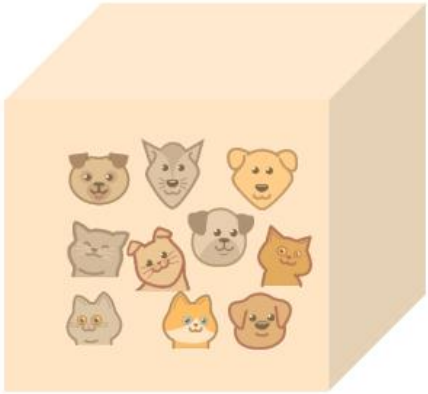

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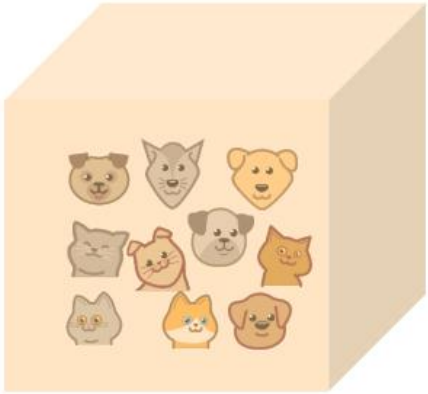

| Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|
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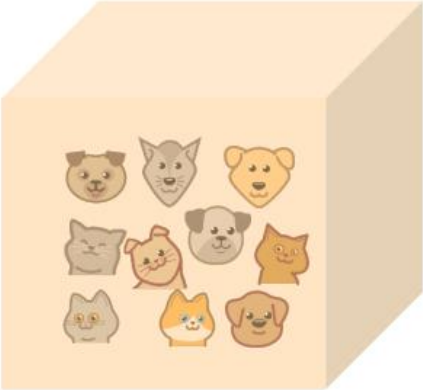
# Sampling with Replacement



| Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|
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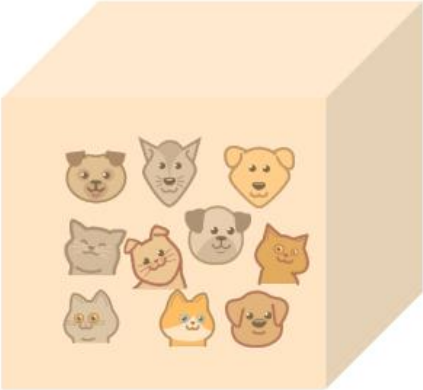
# Sampling with Replacement



| Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|
| Pointy | Round | Present | 1 |
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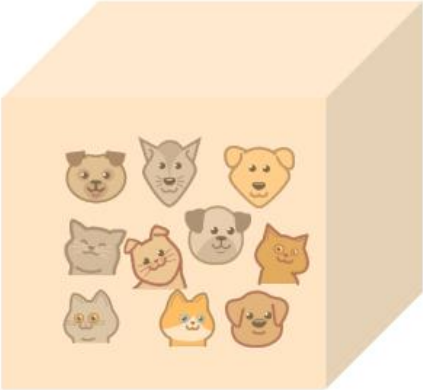
# Sampling with Replacement



| | Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|---|
| | Pointy | Round | Present | 1 |
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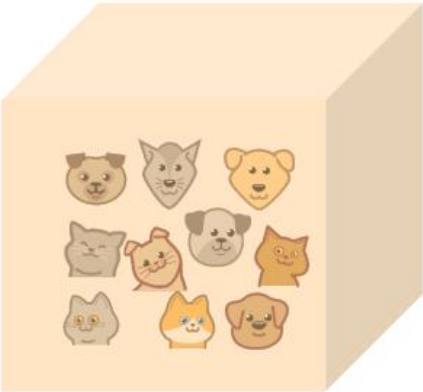

| | Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|---|
| | Pointy | Round | Present | 1 |
| | Floppy | Not round | Absent | 0 |
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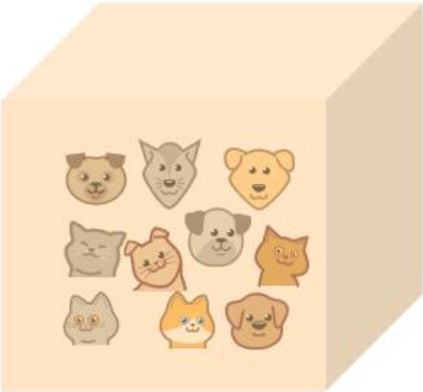

| | Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|---|
| | Pointy | Round | Present | 1 |
| | Floppy | Not round | Absent | 0 |
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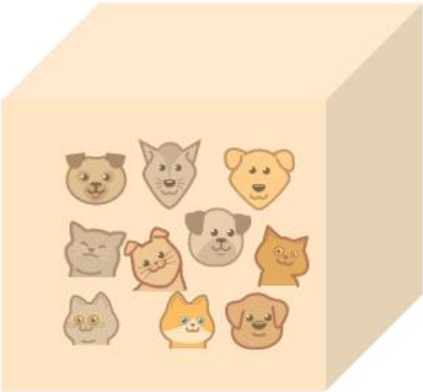
# Sampling with Replacement



| | Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|---|
| | Pointy | Round | Present | 1 |
| | Floppy | Not round | Absent | 0 |
| | Pointy | Round | Absent | 1 |
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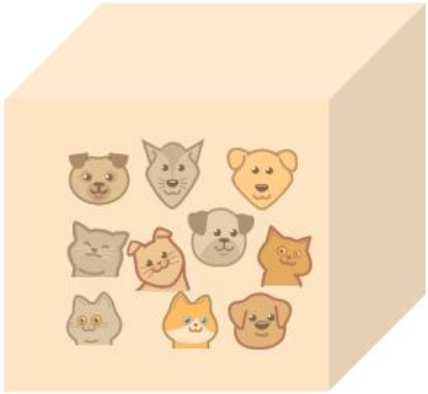

| | Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|---|
| | Pointy | Round | Present | 1 |
| | Floppy | Not round | Absent | 0 |
| | Pointy | Round | Absent | 1 |
| | Pointy | Not round | Present | 0 |
| | Floppy | Not round | Absent | 0 |
| | Pointy | Round | Absent | 1 |
| | Pointy | Round | Present | 1 |
| | Floppy | Not round | Present | 1 |
| | Floppy | Round | Absent | 0 |
| | Pointy | Round | Absent | 1 |

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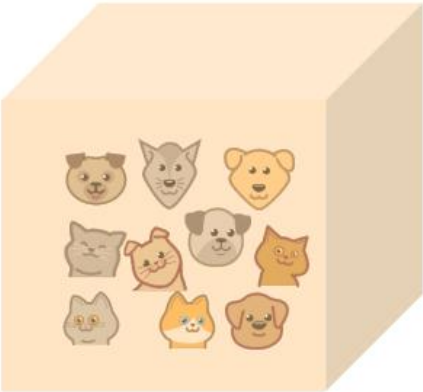
# Sampling with Replacement



| Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|
| Pointy | Round | Present | 1 |
| Floppy | Not round | Absent | 0 |
| Pointy | Round | Absent | 1 |
| Pointy | Not round | Present | 0 |
| Floppy | Not round | Absent | 0 |
| Pointy | Round | Absent | 1 |
| Pointy | Round | Present | 1 |
| Floppy | Not round | Present | 1 |
| Floppy | Round | Absent | 0 |
| Pointy | Round | Absent | 1 |

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# Agenda

- Decision trees
  - Tree ensembles
  - Sampling with replacement
  - **Random forest algorithm**

# Generating Tree Sample

Given training set of size $m$

# Generating Tree Sample

Given training set of size $m$

For $b$ = 1 to $B$:

# Generating Tree Sample

Given training set of size $m$

For $b$ = 1 to $B$:

Use sampling with replacement to create a new training set of size $m$

| Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|
| Pointy | Round | Present | Yes |
| Floppy | Round | Absent | No |
| Floppy | Round | Absent | No |
| Pointy | Round | Present | Yes |
| Pointy | Not Round | Present | Yes |
| Floppy | Round | Absent | No |
| Floppy | Round | Present | Yes |
| Pointy | Not Round | Absent | No |
| Pointy | Not Round | Absent | No |
| Pointy | Not Round | Present | Yes |

# Generating Tree Sample

Given training set of size $m$

For $b$ = 1 to $B$:

Use sampling with replacement to create a new training set of size $m$

Train a decision tree on the new dataset

| Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|
| Pointy | Round | Present | Yes |
| Floppy | Round | Absent | No |
| Floppy | Round | Absent | No |
| Pointy | Round | Present | Yes |
| Pointy | Not Round | Present | Yes |
| Floppy | Round | Absent | No |
| Floppy | Round | Present | Yes |
| Pointy | Not Round | Absent | No |
| Pointy | Not Round | Absent | No |
| Pointy | Not Round | Present | Yes |

# Generating Tree Sample

Given training set of size $m$

For $b$ = 1 to $B$:

Use sampling with replacement to create a new training set of size $m$

Train a decision tree on the new dataset

| Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|
| Pointy | Round | Present | Yes |
| Floppy | Round | Absent | No |
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| Floppy | Round | Absent | No |
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| Pointy | Not Round | Absent | No |
| Pointy | Not Round | Absent | No |
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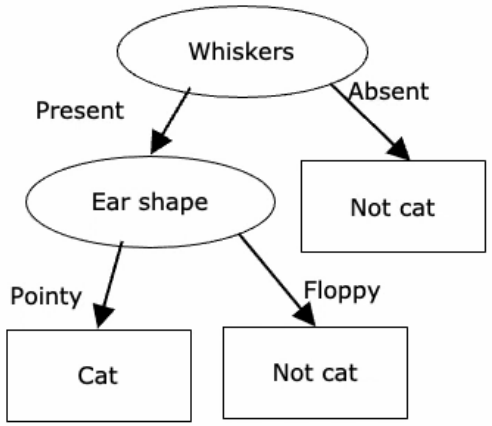


# Generating Tree Sample

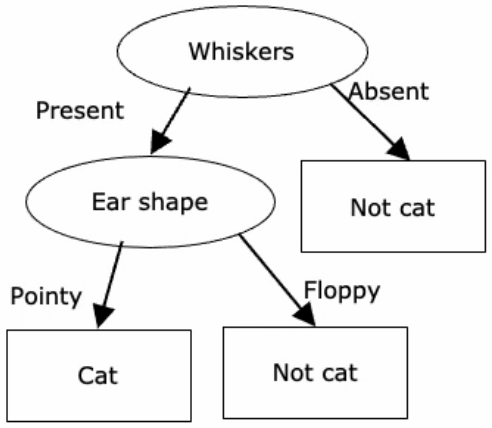
Given training set of size $m$

For $b$ = 1 to $B$:

Use sampling with replacement to create a new training set of size $m$

Train a decision tree on the new dataset

| Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|
| Pointy | Round | Present | Yes |
| Floppy | Round | Absent | No |
| Floppy | Round | Absent | No |
| Pointy | Round | Present | Yes |
| Pointy | Not Round | Present | Yes |
| Floppy | Round | Absent | No |
| Floppy | Round | Present | Yes |
| Pointy | Not Round | Absent | No |
| Pointy | Not Round | Absent | No |
| Pointy | Not Round | Present | Yes |

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|---|---|---|---|
| Pointy | Round | Present | Yes |
| Pointy | Round | Absent | Yes |
| Floppy | Not Round | Absent | No |
| Floppy | Not Round | Absent | No |
| Pointy | Round | Absent | Yes |
| Floppy | Round | Absent | No |
| Floppy | Round | Absent | No |
| Floppy | Round | Absent | No |
| Pointy | Not Round | Absent | No |
| Pointy | Round | Present | Yes |

# Generating Tree Sample

Given training set of size $m$

For $b$ = 1 to $B$:

Use sampling with replacement to create a new training set of size $m$

Train a decision tree on the new dataset

| Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|
| Pointy | Round | Present | Yes |
| Floppy | Round | Absent | No |
| Floppy | Round | Absent | No |
| Pointy | Round | Present | Yes |
| Pointy | Not Round | Present | Yes |
| Floppy | Round | Absent | No |
| Floppy | Round | Present | Yes |
| Pointy | Not Round | Absent | No |
| Pointy | Not Round | Absent | No |
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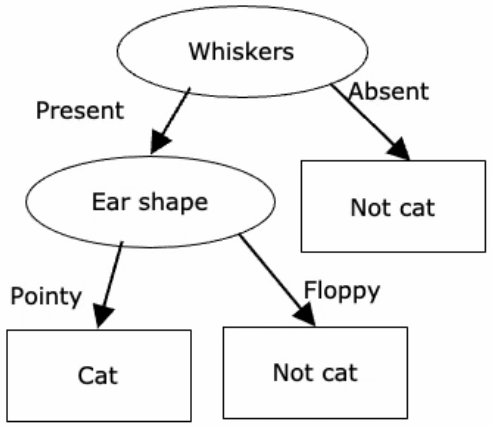


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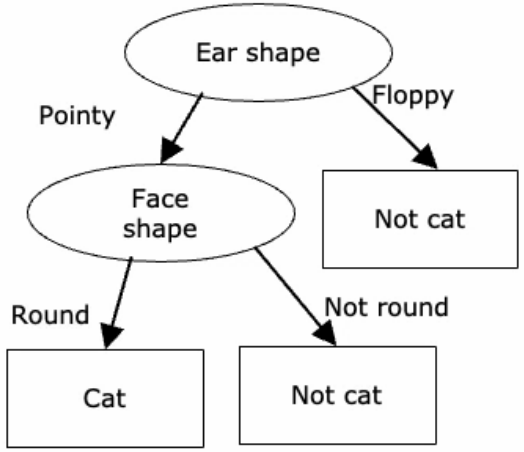
| Pointy | Round | Absent | Yes |
| Floppy | Not Round | Absent | No |
| Floppy | Not Round | Absent | No |
| Pointy | Round | Absent | Yes |
| Floppy | Round | Absent | No |
| Floppy | Round | Absent | No |
| Floppy | Round | Absent | No |
| Pointy | Not Round | Absent | No |
| Pointy | Round | Present | Yes |

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# Generating Tree Sample

Given training set of size $m$

For $b$ = 1 to $B$:

**Number of samples = number of trees trained**

Use sampling with replacement to create a new training set of size $m$

Train a decision tree on the new dataset

| Ear shape | Face shape | Whiskers | Cat |
|---|---|---|---|
| Pointy | Round | Present | Yes |
| Floppy | Round | Absent | No |
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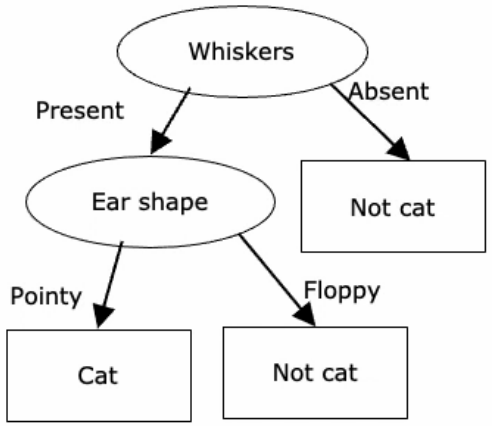


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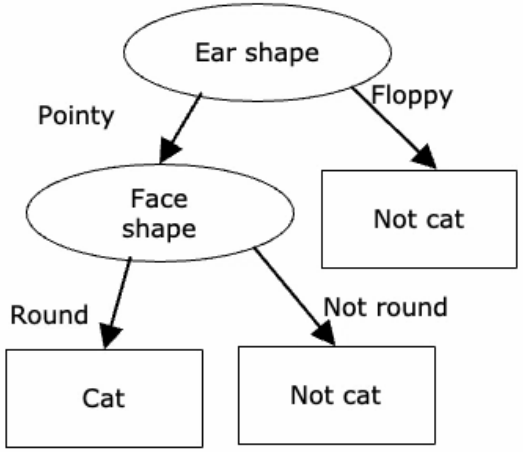

# Generating Tree Sample

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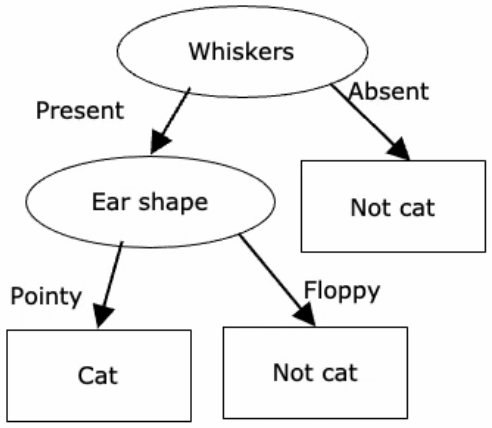


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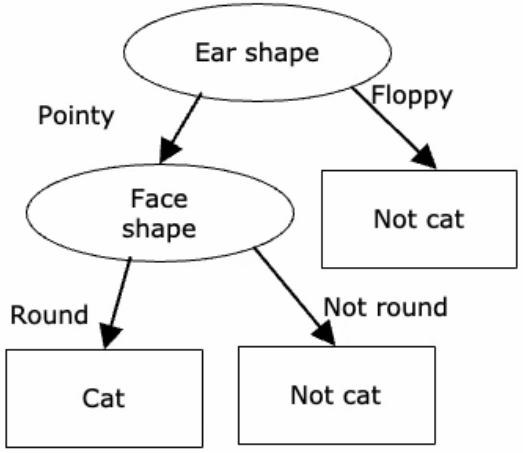


Bagged decision tree

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Plausible number
**64 ~ 128**

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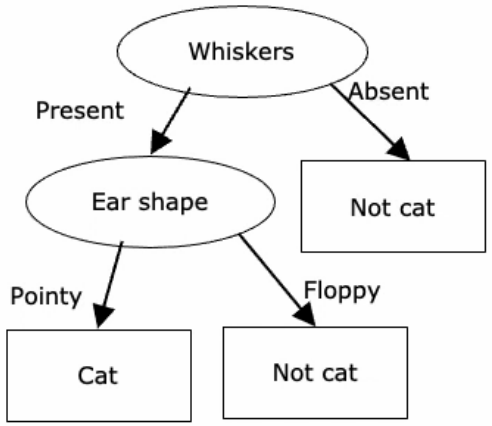


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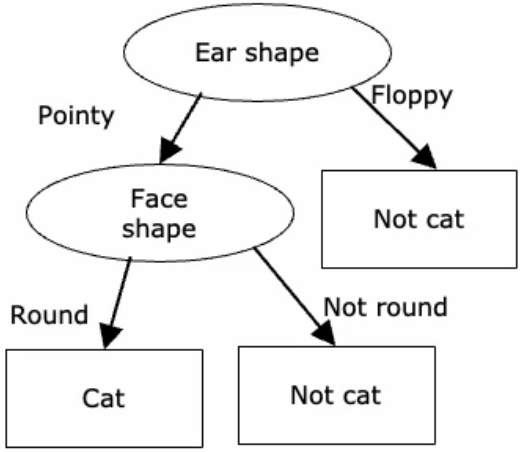


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$$k = \sqrt{n}$$

# Agenda

- Decision trees
  - Tree ensembles
  - Sampling with replacement
  - Random forest algorithm
  - **Boosted trees**

# Boosted Trees

Given training set of size $m$

For $b$ = 1 to $B$:

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Train a decision tree on the new dataset

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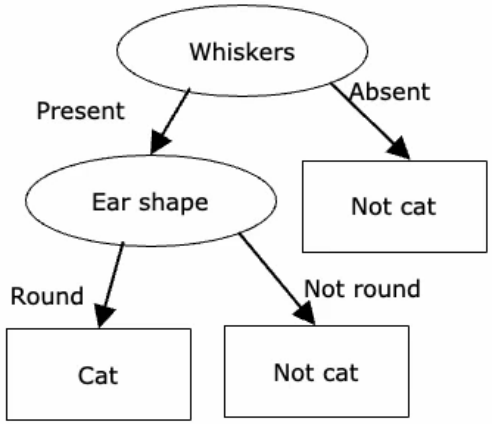

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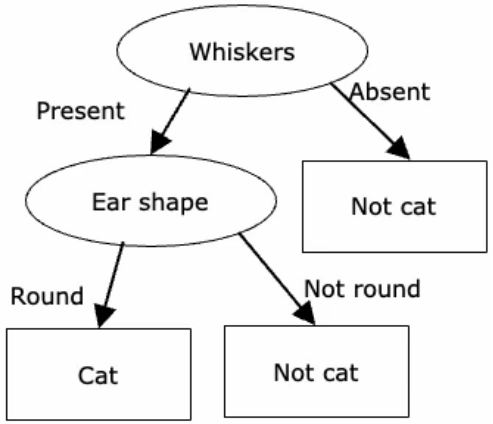


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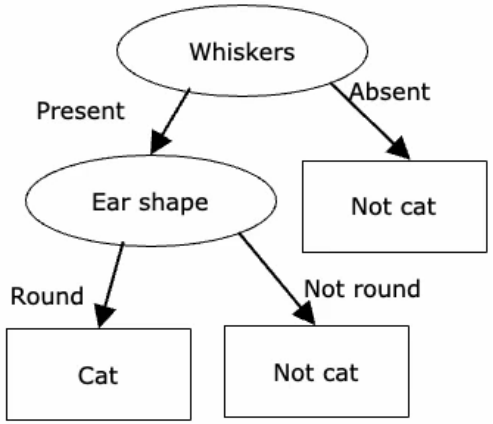


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While training ***b*th** tree, higher probability of selecting training examples misclassified on previous ***b-1*** instances

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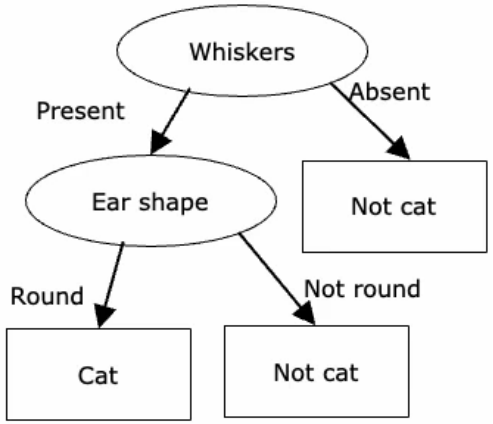


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- **CatBoost:** Suitable for categorical features without one-hot encoding

# Agenda

- Decision trees
  - Tree ensembles
  - Sampling with replacement
  - Random forest algorithm
  - Boosted trees
  - **When to use?**

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  - When building a system of multiple models working together, it might be easier to string together multiple neural networks

# Agenda

- Decision trees
  - Evaluating performance

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- What has not yet been answered is a critical question: how do we know whether a trained model is actually reliable?

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- This balance is at the heart of model **evaluation** and **validation**.

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**Hyperparameters**

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# Training vs Validation vs Test Data

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- For decision trees, complexity is often reflected in how deep the tree is allowed to grow.
  - Shallow trees make only a few splits and capture broad patterns,
  - while deeper trees create many splits and capture fine details.

# Training vs Validation vs Test Data

- We use three subsets and each subset of the data has a specific role.
  - The training data is used to learn patterns and build the structure of the tree or ensemble.
  - The validation data is used to compare different versions of a model and decide which configuration works best.
  - The test data is reserved for a final, unbiased assessment of the chosen model.
- For decision trees, complexity is often reflected in how deep the tree is allowed to grow.
  - Shallow trees make only a few splits and capture broad patterns,
  - while deeper trees create many splits and capture fine details.
- Validation provides a systematic way to compare these alternatives.
  - By training multiple versions of a tree with different levels of complexity and evaluating each one on validation data, we can observe how complexity affects performance on data the model has not seen during training.

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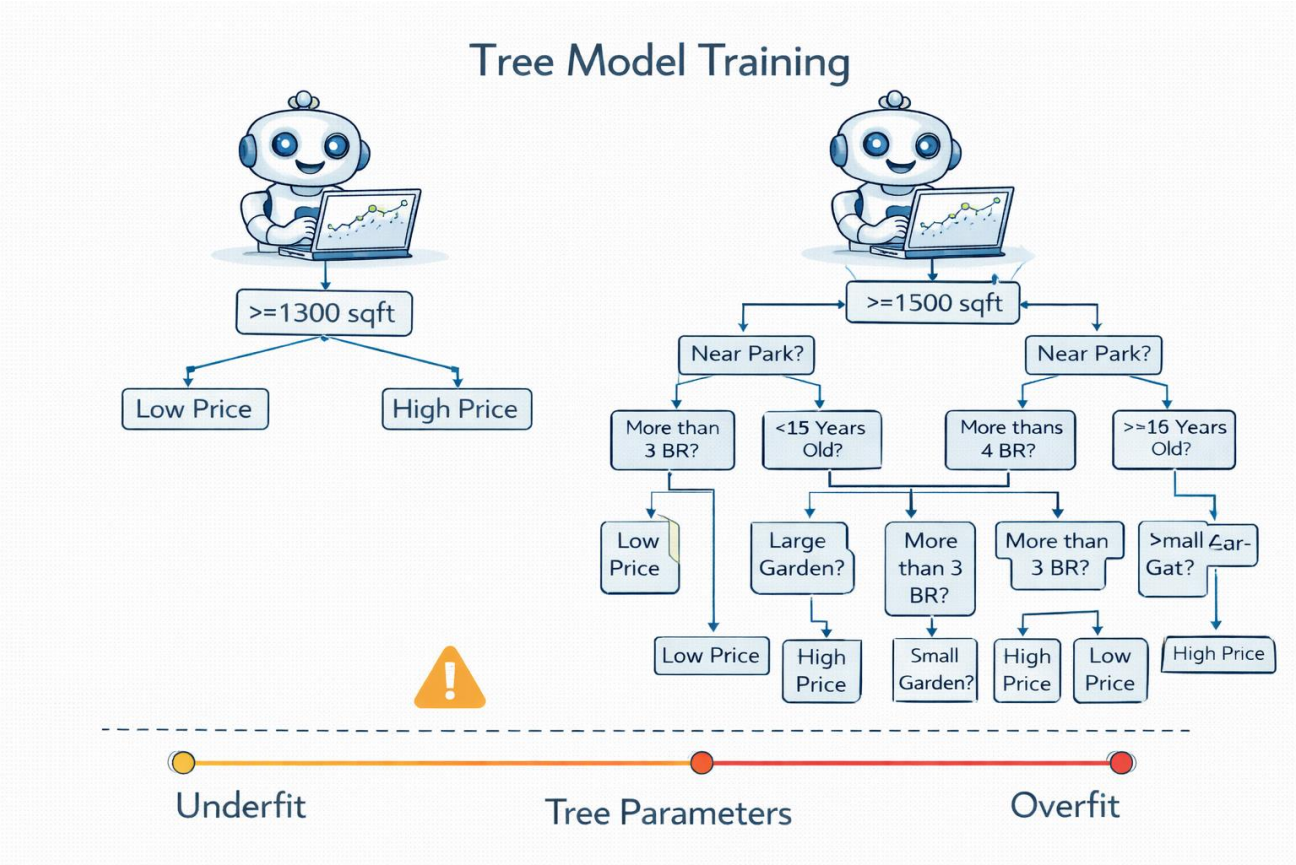
**This process helps identify a balance point where the model is complex enough to capture meaningful structure but not so complex that it starts fitting noise.**

# Validation and Overfitting



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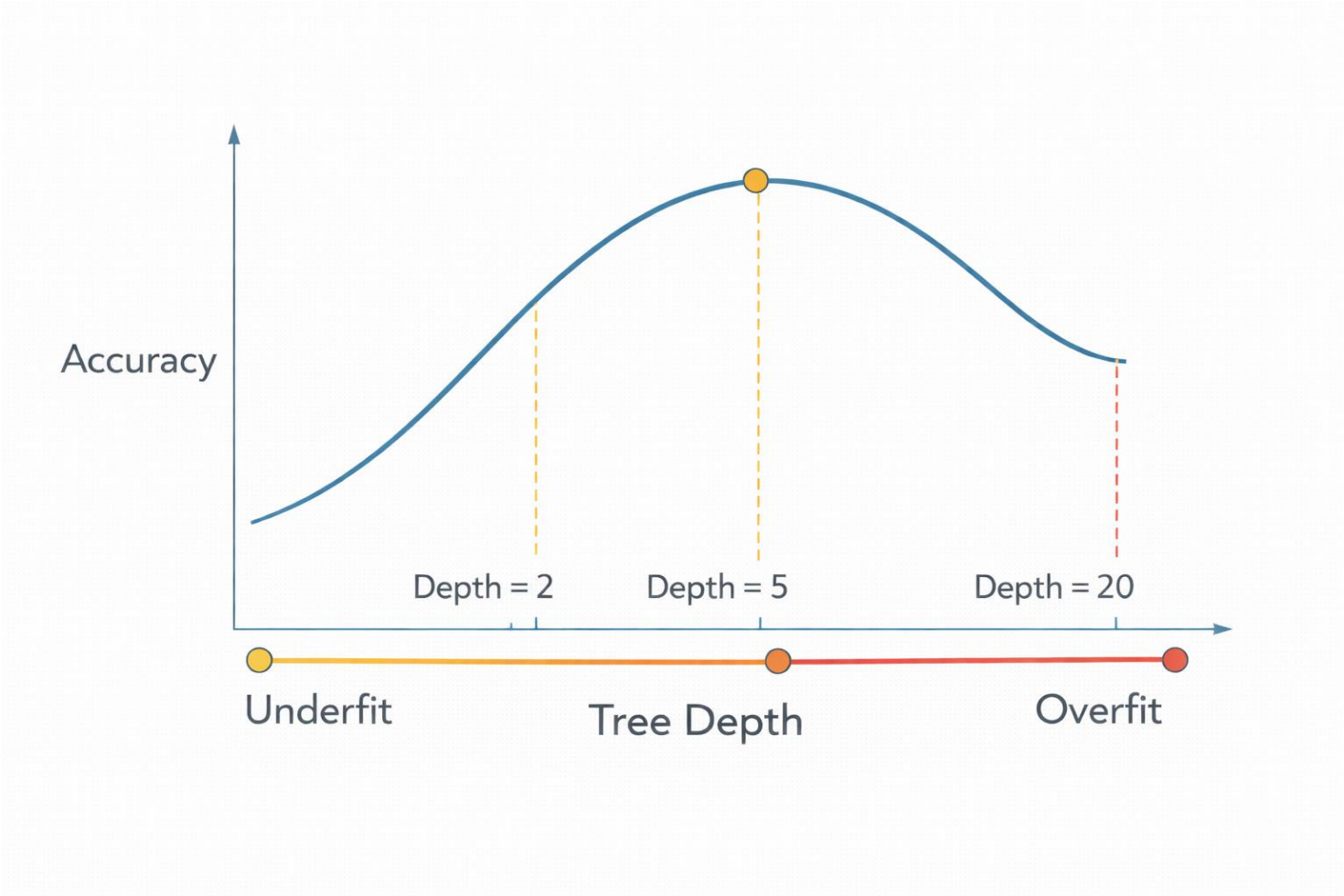

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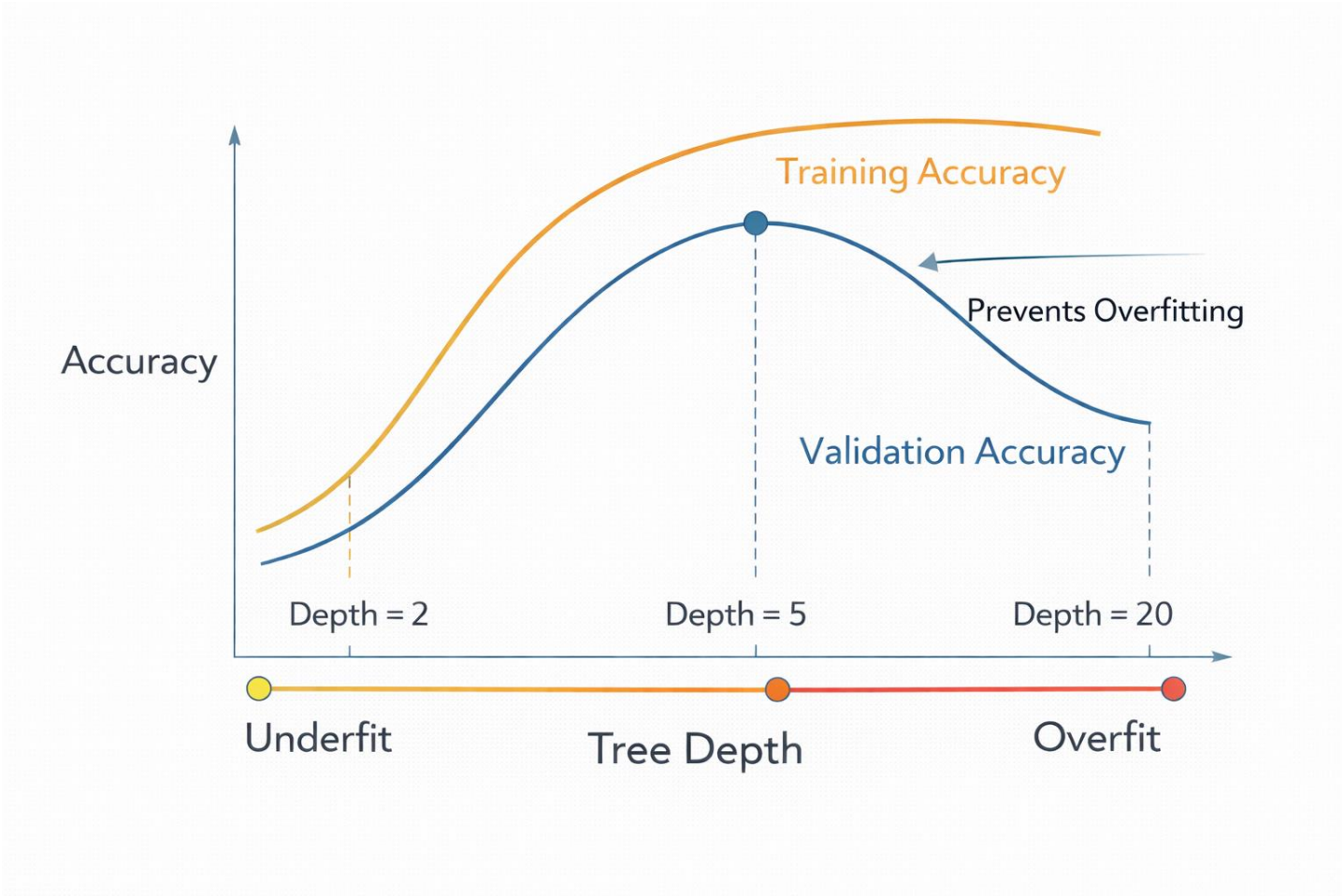

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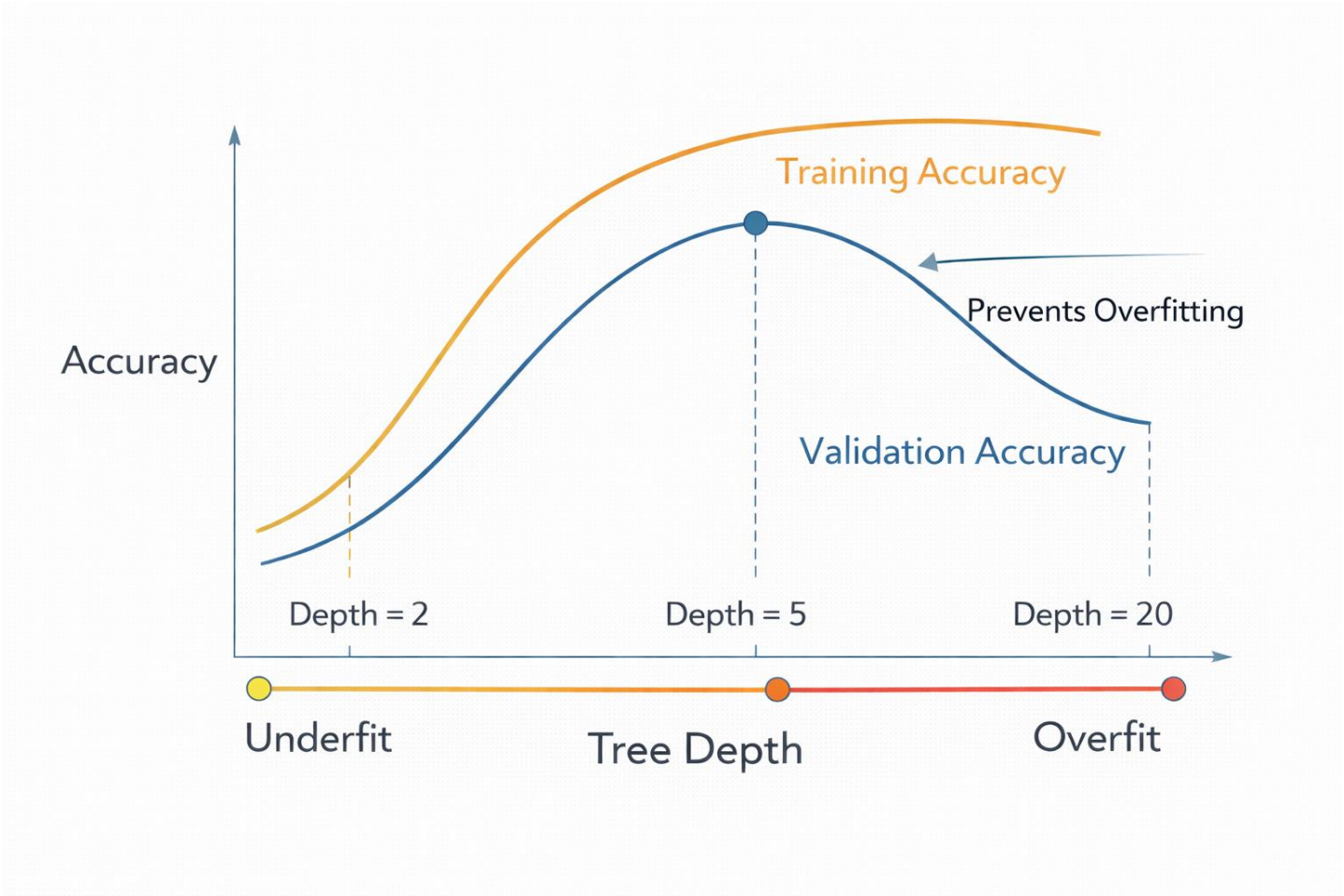


**Bias, Variance Trade-off!**